

Luis Torres, PhD

MEDICAL PHYSICIST · SCIENTIFIC SOLUTIONS ENGINEER

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EXPERIENCE

Flywheel

July 2022 – Present

Senior Scientific Solutions Engineer · Scientific Solutions Engineer

- › Fine-tuned MedSigLIP to classify MRI contrast weighting from pixel data, demonstrating limited-data adaptation of medical foundation models to real-world tasks.
- › Developed a retrieval-augmented generation (RAG) chatbot proof of concept that uses internal documentation to support developers with pipeline design and troubleshooting.
- › Designed and developed automated release-note generation using LLMs in CI/CD pipelines.
- › Lead design and deployment of scalable, containerized imaging workflows across multiple modalities (MRI, CT, PET, ophthalmic).
- › Primary technical partner to industry and academic sites, translating research algorithms and scientific requirements into automated, validated imaging workflows for multi-institution studies, ensuring reproducibility and data integrity across sites.
- › Built modular workflows for image curation, quality control and quantitative analysis across clinical and preclinical datasets.
- › Oversaw ingestion and curation of external imaging datasets, directing contractor teams to ensure data quality, harmonized metadata and reproducibility standards for downstream analysis.
- › Developed and validated modality-specific classification approaches for CT and MR, applying imaging physics to improve labeling consistency, robustness and interpretation.
- › Consulted for imaging scientists, providing hands-on guidance in rule-based classification for MR, CT and PET.
- › Mentored scientific engineers on designing reproducible containerized workflows, development best practices and CI/CD testing strategies.
- › Designed custom Gears for advanced medical image processing, including segmentation, registration and feature extraction.
- › Authored best-practice documentation for workflow containerization.

University of Wisconsin–Madison

August 2016 – June 2022

Graduate Student Researcher

Completed a PhD in Medical Physics focused on quantitative MRI biomarkers and advanced reconstruction methods for pulmonary imaging.

- › Developed and validated semi-automated DCE-MRI and UTE-MRI pipelines, integrating segmentation, motion correction, parametric modeling and advanced registration and denoising, to quantify global and regional microvascular and structural lung changes in idiopathic pulmonary fibrosis and bronchopulmonary dysplasia.
- › Collaborated with clinicians and imaging researchers to interpret imaging-derived biomarkers in the context of pulmonary physiology and disease progression across patient cohorts.

EDUCATION

PhD, Medical Physics

Improved Spatiotemporal Association of Structure and Function in Pulmonary MRI
University of Wisconsin–Madison
Madison, WI · June 2022

MS, Medical Physics

University of Wisconsin–Madison
Madison, WI

BS, Physics

Mathematics minor
New Mexico Institute of Mining and Technology
Socorro, NM · December 2014

SKILLS

Programming

in order of proficiency

Python (NumPy, CuPy, SigPy)
C++ / CUDA
MATLAB
R

Tools

Flywheel
Linux / Unix (bash, tcsh)
Kubernetes / Docker
Git
Slurm / HTCondor / HPC
Advanced Normalization Tools (ANTs)
Berkeley Advanced Reconstruction Toolbox (BART)
ITK

Techniques

Non-linear modeling and regression
Image filtering and denoising
Deconvolution
Image segmentation
Histogram analysis
Non-Cartesian reconstruction
Iterative compressed sensing
Diffeomorphic registration

Los Alamos National Laboratory

February 2015 – August 2016

Post-Baccalaureate Researcher

- › Developed software to process, analyze and visualize multi-sensor environmental imaging datasets, integrating satellite and spectroscopic data for trace gas and atmospheric modeling.

SELECTED PUBLICATIONS

Dynamic contrast enhanced MRI for the evaluation of lung perfusion in idiopathic pulmonary fibrosis

Torres LA, Lee KE, Barton GP, Hahn AD, Sandbo N, Schiebler ML, Fain SB. *European Respiratory Journal*, 2022.

doi:10.1183/13993003.02058-2021

Pulmonary Microvascular Changes in Adult Survivors of Prematurity: Utility of Dynamic Contrast-enhanced Magnetic Resonance Imaging

Barton GP*, Torres LA*, Goss KN, Eldridge MW, Fain SB. *American Journal of Respiratory and Critical Care Medicine*, 2020.

doi:10.1164/rccm.202002-0344LE

Structure-function imaging of lung disease using ultrashort echo time MRI

Torres L, Kammerman J, Hahn AD, Zha W, Nagle SK, Johnson K, Sandbo N, Meyer K, Schiebler M, Fain SB. *Academic Radiology*, 2019.

doi:10.1016/j.acra.2018.12.007

TRAINING

NVIDIA Fundamentals of Deep Learning, DLI certification (2022)
Deep Learning for Medical Imaging Bootcamp (2018)

LANGUAGES

English (native)
Spanish (native)